Grade 8 Name____________________
Unit 4 Assessment Date_________________
No Calculator Period__________

1. Select all of the expressions that are equivalent to $(-8)^{-4} \cdot (-8)^2$.

- ☐ $\frac{1}{64}$
- ☐ $(-8)^{-8}$
- ☐ $(-8)^{-2}$
- ☐ $\frac{(-8)(-8)(-8)(-8)}{(-8)(-8)}$
- ☐ $\frac{(-8)(-8)}{(-8)(-8)(-8)(-8)}$
- ☐ $\frac{1}{(-8)(-8)(-8)(-8)(-8)(-8)(-8)(-8)}$

2. Select all of the expressions that are equivalent to $\left(\frac{1}{6}\right)^{-1}$.

- ☐ $-1\left(\frac{1}{2\cdot 3}\right)^{1}$
- ☐ $-\frac{1}{6}$
- ☐ $\left(\frac{6}{1}\right)^{1}$
- ☐ $\frac{1^{-1}}{6^{-1}}$
- ☐ 6

Rewrite each expression in the form b^x, where x is a positive number.

3. $(-5)^{-5} \cdot (-5)^{9}$

$\boldsymbol{b^x}$ where x is a positive number	

4. $\left(\frac{12}{7}\right)^{-3}$

$\boldsymbol{b^x}$ where x is a positive number	

5. $\left(\left(\frac{6}{11}\right)^{6}\right)^{-4}$

$\boldsymbol{b^x}$ where x is a positive number	

6. $\frac{13^{-8}}{13^{-2}}$

$\boldsymbol{b^x}$ where x is a positive number	

7. $(-18)^{-1}$

$\boldsymbol{b^x}$ where x is a positive number	

8. $(37)^{-24} \cdot (37)^{13}$

$\boldsymbol{b^x}$ where x is a positive number	

9. Which expression is equivalent to $(y)^{-1}$?

A. $-1 \cdot y$

B. $-\frac{1}{y}$

C. y

D. $\frac{1}{y}$

Determine the value of each expression. If necessary, use the workspace to show your work.

10. $\dfrac{(1.1)^{-2} \cdot (1.1)^{3}}{(1.1)^{3} \cdot (1.1)^{-5}}$ **value**	11. $\left(\dfrac{-8}{15}\right)^{-3} \cdot \dfrac{15^{-4}}{(-8)^{-2}}$ **value**
12. $\left(\dfrac{15}{8}\right)^{0} \div \left(\dfrac{2}{5}\right)^{-3}$ **value**	13. $\left(\left(\dfrac{9}{13}\right)^{-2}\right)^{-3} \cdot \left(\dfrac{9}{13}\right)^{-8}$ **value**

For each expression, apply exponent laws to write an equivalent expression with positive exponents that has the fewest factors possible. If necessary, use the workspace to show your work.

14. $(m^8m^3)^2$

equivalent expression	

15. $x^4 \div ((x^2)^3)$

equivalent expression	

16. $\frac{y^5y^{-8}}{y^{-3}y^2}$

equivalent expression	

17. $\left(\frac{2}{w^5} \cdot w^3\right)^{-4}$

equivalent expression	

18. $(4x^4y^5) \cdot (7x^{-8}y^6)$	19. $\left(\frac{a^{-5}n^{13}}{a^{-12}n^{8}}\right)^4$
equivalent expression	**equivalent expression**

20. Select all of the expressions that are equivalent to $\left(\frac{x^{-13}}{x^{11}} \cdot x^{-5}\right)^{-2}$.

- ☐ x^{14}
- ☐ x^{20}
- ☐ $\left(\frac{x^{-13}}{x^{11}} \cdot x^{10}\right)$
- ☐ $\left(\frac{x^{26}}{x^{-22}} \cdot x^{10}\right)$
- ☐ $\left(\frac{1}{x^{24}} \cdot x^{-5}\right)^{-2}$
- ☐ $(x^{-2} \cdot x^{-5})^{-2}$